

Perspective: Consistent resummation schemes for the one-particle Green's function

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Abstract: abstract abstract ...

I. ONE-PARTICLE GREEN'S FUNCTION

As a sum-over-state expression for the interacting one-particle Green's function reads

$$G_{pq}(\omega) = \sum_N \langle \Psi_0 | a_p | \Psi_N \rangle (\omega - \Delta E_N^+)^{-1} \langle \Psi_N | a_q^\dagger | \Psi_0 \rangle + \sum_M \langle \Psi_0 | a_q^\dagger | \Psi_M \rangle (\omega - \Delta E_M^-)^{-1} \langle \Psi_M | a_p | \Psi_0 \rangle \quad (1)$$

with $a_q^\dagger$ and a_p denoting fermionic creation and annihilation operators, and $\Delta E_N^+ = E_N^+ - E_0$ and $\Delta E_M^- = E_0 - E_M^-$. With + and - referring to the energy of system with an added/removed electron.

As an alternative to Eq. (1), the one-particle Green's function can be determined through the Dyson equation

$$G_{pq}(\omega) = G_{pq}^0(\omega) + \sum_{rs} G_{pr}^0(\omega) \Sigma_{rs}(\omega) G_{sq}(\omega), \quad (2)$$

with $G_{pq}^0(\omega)$ denoting the non-interacting Green's function

$$G_{pq}^0(\omega) = \delta_{pq} \frac{1 - n_p}{\omega + \epsilon_p} + \delta_{pq} \frac{n_p}{\omega - \epsilon_p}, \quad (3)$$

with n_p denoting the occupation number from the Fermi-Dirac distribution. We assume the zero temperature limit throughout, n_p associated with a mean-field orbital p is therefore either 0 (for an unoccupied orbital) or 1 (for an occupied orbital). Eq. (2) can also be rearranged to (in explicit matrix form)

$$\mathbf{G}(\omega) = \frac{1}{\mathbf{G}_0^{-1}(\omega) - \mathbf{\Sigma}(\omega)}. \quad (4)$$

The connection between the sum-over-state expression and the Dyson equation is made explicit in the following. For this, we define a new vacuum $|\Psi_0\rangle$

$$|\Psi_0\rangle = e^\sigma |0\rangle, \quad (5)$$

via exponential parametrization of the reference mean-field state $|0\rangle$. σ denotes an anti-Hermitian particle-hole excitation operator

$$\sigma = \hat{T} - \hat{T}^\dagger, \quad (6)$$

with

$$\hat{T} = \sum_v t_v \bar{\tau}_v. \quad (7)$$

$\bar{\tau}_v$ represents an arbitrary particle-hole excitation operator ($\bar{a}_a^\dagger \bar{a}_i + \dots$). Note that the amplitudes t_v [Eq. (7)] are determined by projection and satisfy the general Brioullin conditions

$$\langle \Psi_0 | [\hat{H}, \bar{\tau}_v - \bar{\tau}_v^\dagger] | \Psi_0 \rangle = 0. \quad (8)$$

In the following, occupied orbitals are referred to with subscripts $i, j, k, \dots$, virtual orbitals with $a, b, c, \dots$ and arbitrary orbitals with $p, q, r, \dots$. Furthermore, the particle and hole creation and annihilation operators acting on the vacuum are (still obeying their anti-commutation relations)

$$\bar{a}_p = e^\sigma a_p e^{-\sigma} \quad (9)$$

$$\bar{a}_p^\dagger = e^\sigma a_p^\dagger e^{-\sigma}. \quad (10)$$

Within this formalism an arbitrary excited state N is obtained as

$$|\Psi_N\rangle = \hat{R}_N |\Psi_0\rangle. \quad (11)$$

$\hat{R}_N$ is given as

$$\begin{aligned} \hat{R}_N |\Psi_0\rangle &= \sum_v R_v^N \bar{\tau}_v |\Psi_0\rangle \\ &= \sum_v R_v^N e^\sigma \hat{\tau}_v e^{-\sigma} |\Psi_0\rangle = \sum_v R_v^N e^\sigma |\Phi_v\rangle, \end{aligned} \quad (12)$$

with R_v^N denoting the expansion coefficients so that

$$|\Psi_N\rangle = \sum_v R_v^N e^\sigma \hat{\tau}_v |0\rangle. \quad (13)$$

Note that $\hat{\tau}_v$ is chosen according to the excited state of interest, e.g., for charged ($N \pm 1$) excitations

$$\hat{\tau}_v^- = \{a_i, a_b^\dagger a_j, \dots\} \quad (14)$$

$$\hat{\tau}_v^+ = \{a_a^\dagger, a_a^\dagger a_b^\dagger a_j, \dots\} \quad (15)$$

$$\hat{\tau}_v = \{\hat{\tau}_v^-\} \cup \{\hat{\tau}_v^{+, \dagger}\} \quad (16)$$

so that

$$\begin{aligned} 1 &= \sum_v \bar{\tau}_v |\Psi_0\rangle \langle \Psi_0 | \bar{\tau}_v^\dagger \\ 1 &= \sum_v e^\sigma \hat{\tau}_v |0\rangle \langle 0 | \hat{\tau}_v^\dagger e^{-\sigma} \end{aligned} \quad (17)$$

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is satisfied. The Hamiltonian for that new vacuum is defined in terms of the new set of creation and annihilation operators

$$\begin{aligned}\bar{H} &= e^\sigma \hat{H} e^{-\sigma} \\ &= \sum_{pq} F_{pq} \bar{a}_p^\dagger \bar{a}_q + \frac{1}{4} \sum_{pqrs} V_{pqsr} \bar{a}_p^\dagger \bar{a}_q^\dagger \bar{a}_r \bar{a}_s.\end{aligned}\quad (18)$$

Within the newly defined quantities excitation energy differences ΔE_N are evaluated using the EOM approach

$$\begin{aligned}\langle \Psi_0 | [\hat{r}_v^\dagger, [\bar{H}, \sum_\mu R_\mu^N \bar{r}_\mu]] | \Psi_0 \rangle \\ = \Delta E_N \langle \Psi_0 | [\hat{r}_v^\dagger, \sum_\mu R_\mu^N \bar{r}_\mu] | \Psi_0 \rangle,\end{aligned}\quad (19)$$

which can be rewritten to

$$\begin{aligned}\langle 0 | [\hat{r}_v^\dagger, [\hat{H}, \sum_\mu R_\mu^N \hat{r}_\mu]] | 0 \rangle \\ = \langle 0 | [\hat{r}_v^\dagger, \sum_\mu R_\mu^N \hat{r}_\mu] | 0 \rangle \Delta E_N.\end{aligned}\quad (20)$$

Eq. (20) resembles the common EOM configuration interaction approach and can be explicitly written in matrix form

$$\mathbf{H}\mathbf{R} = \mathbf{S}\mathbf{R}\mathbf{E} \quad (21)$$

$$\Leftrightarrow \tilde{\mathbf{H}}\mathbf{R} = \mathbf{R}\mathbf{E} \quad (22)$$

(assuming $\mathbf{S}$ is invertible and a symmetric transformation). With the above definitions, Eq. (1) reads

$$\begin{aligned}G_{pq}(\omega) &= \sum_N \langle \Psi_0 | \bar{a}_p | \Psi_N \rangle (\omega - \Delta E_N^+)^{-1} \langle \Psi_N | \bar{a}_q^\dagger | \Psi_0 \rangle \\ &+ \sum_M \langle \Psi_0 | \bar{a}_q^\dagger | \Psi_M \rangle (\omega - \Delta E_M^-)^{-1} \langle \Psi_M | \bar{a}_p | \Psi_0 \rangle,\end{aligned}\quad (23)$$

which can be rearranged to

$$\mathbf{G}_{11}(\omega) = \begin{pmatrix} \mathbf{1}_{11} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_{11} \\ \mathbf{0} \end{pmatrix}, \quad (24)$$

where the subscript 11 denotes the p/h subspace and 2 refers to all higher order excitations, i.e., hhp, pph etc.. The connection to the Dyson equation becomes apparent by rewriting Eq. (24)

$$\begin{aligned}\mathbf{G}_{11}(\omega) &= [\omega \mathbf{1}_{11} - \mathbf{H}_{11} - \mathbf{H}_{12} [\omega \mathbf{1}_{22} - \mathbf{H}_{22}]^{-1} \mathbf{H}_{21}]^{-1} \\ \Rightarrow \mathbf{G}(\omega) &= [\omega - \mathbf{F} - \Sigma(\infty) - \tilde{\Sigma}(\omega)]^{-1} \\ \Leftrightarrow \mathbf{G}(\omega) &= [\omega - \mathbf{F} - \Sigma(\omega)]^{-1} \\ \Leftrightarrow \mathbf{G}(\omega) &= [\mathbf{G}_0^{-1}(\omega) - \Sigma(\omega)]^{-1}.\end{aligned}\quad (25)$$

Consequently, we have demonstrated the connection of the effective Hamiltonian used for the determination of single-particle excitation energies and the Dyson equation of the one-particle Green's function. For generality, we have partitioned H_{11} into the bare Fock matrix and additional screening effects summarized in $\Sigma(\infty)$ (for example HF exchange is often

considered to be part of $\Sigma(\infty)$). Furthermore, particle and hole space are now intrinsically coupled in $\Sigma(\omega)$.

The route described above is not the strategy commonly used in quantum chemistry for the determination of the one-particle Green's function. Instead, the original definition of creation and annihilation operators is kept, so that the EOM reads

$$\begin{aligned}\langle 0 | [\hat{r}_v^\dagger, [\bar{H}, \sum_\mu R_\mu^N \hat{r}_\mu]] | 0 \rangle \\ = \langle 0 | [\hat{r}_v^\dagger, \sum_\mu R_\mu^N \hat{r}_\mu] | 0 \rangle \Delta E_N,\end{aligned}\quad (26)$$

with $\bar{H}$ denoting the unitary transformed Hamiltonian

$$\bar{H} = e^{-\sigma} \hat{H} e^\sigma \quad (27)$$

and the one-particle Green's function written as

$$\begin{aligned}G_{pq}(\omega) &= \sum_{v,\mu} \langle 0 | e^{-\sigma} a_p e^\sigma | \Phi_v \rangle [(\omega - \mathbf{H}^+)^{-1}]_{v\mu} \langle \Phi_\mu | e^{-\sigma} a_q^\dagger e^\sigma | \Psi_0 \rangle \\ &+ \sum_{v,\mu} \langle 0 | e^{-\sigma} a_q^\dagger e^\sigma | \Phi_v \rangle [(\omega - \mathbf{H}^-)^{-1}]_{v\mu} \langle \Phi_\mu | e^{-\sigma} a_p e^\sigma | \Psi_0 \rangle,\end{aligned}\quad (28)$$

with $\mathbf{H}^+$, $\mathbf{H}^-$ denoting the EOM Hamiltonian in the complete $N \pm 1$ excitation basis. Note that no coupling between these two sectors occurs due to Eq. (8). Furthermore, Φ_v denotes excited Slater determinants in the $N \pm 1$ sectors [compare Eq. (12)]. Next, we write Eq. (28) compactly as

$$\begin{aligned}\mathbf{G}_{11}(\omega) &= \\ &= \begin{pmatrix} \mathbf{M}_{11}^\dagger & \mathbf{M}_{21}^\dagger \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{M}_{11} \\ \mathbf{M}_{21} \end{pmatrix}\end{aligned}\quad (29)$$

(note that in the blocks the $N+1$ and $N-1$ are still decoupled). $\mathbf{M}_{11}$ denotes contributions from $\langle \Phi_\mu | e^{-\sigma} a_q^\dagger e^\sigma | \Psi_0 \rangle$ and $\langle \Phi_\mu | e^{-\sigma} a_p e^\sigma | \Psi_0 \rangle$ in the 1h-1p sector and $\mathbf{M}_{12}$ in all higher excitation sectors. It becomes clear that the terms such as $\langle 0 | e^{-\sigma} a_p e^\sigma | \Phi_v \rangle$ in Eq. (28) prohibit a simple projection of the decoupled EOM Hamiltonians $\mathbf{H}^+/\mathbf{H}^-$ in the 1-h/1-p space as in Eq. (24). Consequently the precise connection to the Dyson equation can be only made approximately by separating $e^{-\sigma} a_q^\dagger e^\sigma$ into zeros order term a_q and higher order terms of the BCH. Consequently, Eq. (29) is rearranged to

$$\begin{aligned}\mathbf{G}_{11}(\omega) &= \\ &= \begin{pmatrix} \mathbf{1}_{11} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{1}_{11} \\ \mathbf{0} \end{pmatrix} \\ &+ \begin{pmatrix} \tilde{\mathbf{M}}_{11}^\dagger & \mathbf{M}_{21}^\dagger \end{pmatrix} \begin{pmatrix} \omega \mathbf{1}_{11} - \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \omega \mathbf{1}_{22} - \mathbf{H}_{22} \end{pmatrix}^{-1} \begin{pmatrix} \tilde{\mathbf{M}}_{11} \\ \mathbf{M}_{21} \end{pmatrix}.\end{aligned}\quad (30)$$

From the above considerations for the one-particle Green's function it becomes apparent that a ground-state p/h transformation of the Hamiltonian $\hat{H}$ prohibits the determination of $\mathbf{G}$ through the Dyson equation [Eq. (2)] exactly. Instead, the different components of $\mathbf{G}$ have to be determined separately. (See also the work by Tarantelli and Cederbaum¹).

REFERENCES

- ¹A. Tarantelli and L. Cederbaum, “Configuration-interaction formulation of the dyson equation,” *Phys. Rev. A* **45**, 2790 (1992).